AgroBot: Universal AI-Based Multilingual Agricultural Assistant

1. Introduction

Agriculture remains a critical sector globally, yet many farmers lack access to timely expert guidance. AgroBot is an AI-powered solution that provides real-time crop disease detection, multilingual communication, and intelligent agricultural insights.

2. Problem Statement

Farmers face challenges such as difficulty in identifying crop diseases early, limited access to experts, language barriers, and inefficient planning, leading to crop losses and reduced productivity.

3. Proposed Solution

AgroBot integrates AI technologies to assist farmers with:

- Crop disease detection using images

- Multilingual chatbot support

- Weather-based advice

- Crop planning tools

- Market price insights

4. System Modules

AI Chat Module: Multilingual support and instant advice

Image Analysis Module: Disease detection and scoring

Crop Planner Module: Scheduling and reminders

Weather Module: Forecast and recommendations

Market Intelligence Module: Price trends and alerts

Community Module: Farmer interaction and discussions

5. Methodology

- Data Collection

- Data Processing

- Model Implementation

- Integration

- User Interface Design

6. Technologies Used

Frontend: HTML, CSS, JavaScript

Backend: Java / Spring Boot

Database: MongoDB

AI/ML: Image recognition & NLP

APIs: Weather and Market APIs

7. Results & Output

The system detects diseases, provides recommendations, and displays insights effectively.

8. Advantages

- Reduced crop losses

- Less dependency on experts

- Improved productivity

- Multilingual support

- Scalable solution

9. Limitations

- Requires internet

- Depends on image quality

- Needs model updates

10. Future Enhancements

- Offline mode

- IoT integration

- Predictive analytics

- Voice interaction

11. Conclusion

AgroBot bridges traditional farming and AI, improving decision-making and productivity.